

MONCHEGORSK, Russian Federation

WMO: 222120

Lat: 67.9422N

Lon: 32.9242E

Elev: 432

StdP: 14.47

Time Zone: 3.00 (E03)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-15.8	-29.6	1.1	-23.4	-19.8	1.9	-15.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	76.5	61.5	72.7	60.0	69.0	57.9	64.1	74.4	61.3	69.5	59.1	66.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.8	78.0	67.6	57.4	71.5	65.1	55.6	66.8	62.6	29.5	75.2	27.5	70.6	25.9	66.9	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-28.4	81.1	4.7	4.2	-31.8	84.1	-34.5	86.6	-37.2	88.9	-40.6	92.0
			-28.5	65.0	4.7	3.3	-31.9	67.3	-34.7	69.3	-37.3	71.1	-40.8	73.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.8	9.5	16.1	21.2	30.3	42.6	50.6	58.7	55.1	46.6	34.6	23.6	15.6
	DBStd	18.63	14.14	11.39	10.26	7.41	8.60	7.80	7.94	6.27	5.91	6.88	9.59	12.00
	HDD50	6537	1256	948	894	590	262	84	13	16	135	479	793	1067
	HDD65	11434	1721	1368	1359	1040	695	437	228	315	552	944	1243	1532
	CDD50	627	0	0	0	0	34	101	283	176	33	0	0	0
	CDD65	47	0	0	0	0	1	5	32	9	0	0	0	0
	CDH74	219	0	0	0	0	7	29	145	38	0	0	0	0
	CDH80	20	0	0	0	0	0	1	16	3	0	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	21.8	1.5	1.3	1.1	1.1	1.7	2.2	2.9	2.4	2.2	2.0	1.7	1.7
	PrecMax	25.5	2.8	2.4	2.1	2.6	3.2	4.4	5.4	4.8	4.2	4.1	3.5	2.4
	PrecMin	15.3	0.4	0.6	0.5	0.3	0.6	0.6	1.2	0.9	0.7	0.6	0.6	0.5
	PrecStd	2.6	0.6	0.5	0.4	0.5	0.7	0.8	0.9	0.9	0.9	0.8	0.7	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	39.3	42.2	52.6	73.4	78.3	82.2	78.8	63.5	51.1	41.4	38.1
		MCWB	35.5	35.4	35.1	40.8	54.2	60.3	65.5	63.0	54.6	47.3	37.2	34.5
	2%	DB	34.9	35.7	38.7	45.7	65.6	71.0	77.9	73.1	59.8	48.0	38.2	34.9
		MCWB	32.3	32.8	33.2	38.1	51.3	57.1	62.3	61.6	53.0	44.1	36.1	32.5
	5%	DB	29.5	33.4	36.4	42.2	61.1	66.9	74.7	67.7	57.1	45.8	35.8	32.4
		MCWB	28.0	31.5	32.4	36.7	48.3	54.3	60.9	58.7	52.2	42.7	34.4	30.5
	10%	DB	26.2	31.3	33.9	39.2	56.0	62.9	71.6	64.1	54.6	43.6	34.2	30.3
		MCWB	25.1	29.4	30.8	35.0	46.4	52.8	59.9	57.5	50.6	41.1	32.8	28.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	35.8	35.8	41.5	55.5	63.4	67.5	67.9	56.6	48.1	38.1	35.0
		MCDB	38.8	38.5	40.7	50.0	69.1	75.0	76.9	74.4	60.0	50.0	40.3	37.3
	2%	WB	31.5	33.5	34.1	38.8	52.5	58.4	65.8	62.3	54.5	44.7	36.4	32.9
		MCDB	34.1	35.5	37.8	44.4	64.0	67.3	75.5	70.9	58.2	47.1	38.0	34.8
	5%	WB	27.7	31.8	32.7	37.0	49.8	55.8	62.9	59.5	52.5	43.0	34.7	30.9
		MCDB	29.1	33.4	35.8	41.1	59.9	63.3	71.4	65.5	56.5	45.3	35.9	32.3
	10%	WB	25.2	29.2	31.1	35.7	46.6	53.7	60.5	57.8	50.9	41.3	33.1	28.8
		MCDB	26.2	31.1	33.7	38.8	55.0	61.9	68.8	63.4	54.4	43.4	34.1	30.3
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	10.1	11.3	10.8	11.9	11.0	11.6	8.8	7.8	5.8	8.1	11.2
		MCDBR	11.0	8.3	8.3	13.4	20.7	17.3	17.6	14.2	11.9	8.5	7.4	9.9
	5% WB	MCWBR	9.9	7.0	6.0	8.1	10.4	7.8	6.6	5.9	6.5	6.5	6.6	8.8
		MCWBR	10.8	9.0	7.9	11.3	19.5	14.3	14.8	11.6	10.9	7.0	7.3	9.5
(40) Clear-Sky Solar Irradiance	taub	taub	0.209	0.264	0.305	0.340	0.359	0.357	0.375	0.356	0.323	0.283	N/A	N/A
		taud	1.908	2.041	2.078	2.125	2.294	2.424	2.403	2.431	2.488	2.244	N/A	N/A
	Ebn at Noon	Ebn at Noon	28	162	225	252	263	268	258	250	230	160	N/A	0
		Edh at Noon	3	16	28	36	34	31	30	27	20	13	N/A	0
(44) All-Sky Solar Radiation	RadAvg	RadAvg	12	147	582	1116	1430	1558	1430	987	539	168	28	0
	RadStd	RadStd	1	21	47	80	136	146	184	96	51	24	3	0

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page